

An intelligent feature selection method using binary teaching-learning based optimization algorithm and ANN

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ABSTRACT

The most challenging issue in dealing with big datasets is the large number of their dimensions. Feature selection is a technique for reducing the dimensionality of datasets by removing irrelevant and useless features from the dataset, finally selecting the most crucial features in the dataset, will ultimately increase the algorithm's efficiency. This paper presents a novel feature selection procedure that includes a binary-modified teaching learning-based optimization algorithm (BMTLBO). One of the most effective and popular methods is the TLBO algorithm. Although this algorithm has a good convergence rate, there is a chance to get stuck in a local optimum. Our goal is to balance between exploration and exploitation capabilities. The proposed method consists of two parts: First, the BMTLBO algorithm, which includes the improved binary version of the basic algorithm, is used for the feature selection problem. The idea of a pool is used in this algorithm that boosts the population's diversity while enhancing the algorithm's precision and convergence rate. Second, we are used from improved TLBO algorithm with the self-learning phase (SLTLBO) for neural network training to show the application of the classification problem to evaluate the performance of the procedures of the method. We are tested the proposed method on 14 datasets in the term of classification accuracy and the number of features. Additionally, a different chemical data set named Biodegradable is used to assess the proposed method. The evaluation results demonstrate that, in terms of accuracy, convergence rate, and success in achieving promising solutions, the proposed algorithm outperforms all other optimization algorithms compared. The results are very promising and close to optimal.

1. Introducing

The idea behind data mining is to find hidden information or particular patterns and relationships among a large amount of data in one or more large datasets. Data mining is used to find and make use of the vast amounts of datasets that exist. Each big data has several features, and the set of features describe the situation of each data record. One of the biggest problems is dimensionality, and working with a data set that has a large number of dimensions can make the algorithm take longer and be more computationally complex, big data has a problem with extra, unnecessary features that were added to the original data set in addition to having a lot of features. Some datasets may also contain noise. The vast number of features is too much for traditional data mining techniques to handle, they may stuck in a local optimum. Pre-processing of big data is necessary to remove redundant, unnecessary, and extraneous features and to reduce the dimensions of the data. This preprocessing phase is called feature selection. The tool of feature

selection in data mining is one of the most significant and well-known. The process of identifying pertinent features and eliminating unnecessary and redundant features is referred to as feature selection. The main goal of feature selection is to find the best set of big data features with the highest level of classification accuracy.

Three classes of feature selection techniques are typically offered: filter method, embedded method and wrapper method. The filter method filters features based on characteristics of a dataset that are common, such as correlation with the dependent variable. Without using a predictive model, filtering is done. When there are a lot of features, this method is usually quicker and more effective [1]. While avoiding over fitting, it occasionally may fail to choose the best features. Because feature selection is integrated into the learning or model-building phases in embedded classes, there is a lower chance of over-fitting in this category, and finally, the category of wrapper method finds the best features along with the prediction model and uses the same model to save best features [2]. Better performance is achieved

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despite the computational expense and over-fitting risk. Over fitting is an undesirable machine learning behavior that occurs when the machine learning model provides accurate predictions for the training data but not for the unknown data. Machine learning models are first trained on a known data set by data before they use them to make predictions. The model then tries to predict the outcomes for the new data set using this knowledge. Therefore, the higher over fitting value, the lower the prediction accuracy of the unknown data and the higher the classification error. According to previous research, evolutionary algorithms have been shown to perform more effectively than traditional algorithms. In the following, we will introduce research for each of these three categories and examine them. A filter-based feature selection algorithm is an ant colony optimization (ACO) for feature selection [3]. In another study, the ACO algorithm was employed to choose redundancy-related features (RRFSACO) [4]. For feature selection issues, genetic algorithms are frequently used, for example, the embedded feature selection method based genetic programming (GP) is presented [5] and a hybrid method includes a genetic algorithm (HGAWF) [6,7]. In the wrapper feature selection method, evolutionary algorithms such as a binary version of water wave optimization (BWFO) are used directly [8], altruistic whale optimization algorithm (AITWOA) [9], chaotic binary group search optimizer (CGSO) [10] and A Feature selection technique based on a novel equilibrium Optimization is presented [11]. Light GBM is also a wrapper-based feature selection method that consists of the gradient-boosting framework which uses a tree-based learning algorithm. The various protein structures have been categorized using this algorithm. In particular, the Phage_UniR_LGBM method is suggested to categorize virion proteins. Such a model makes use of the UniRep feature and the LightGBM algorithm as a classification model [12].

In this paper evolutionary algorithms are presented with two goals: minimize the number of features and maximize classification accuracy, hence we used two different versions of the TLBO algorithm [13]. TLBO algorithm draws its inspiration from the way teachers teach and how students pick up knowledge from both the teacher and one another. The use of TLBO in numerous applications demonstrated its superiority. This algorithm has grown to be one of the most well-liked optimization algorithms as a result of its benefits, including strong exploration and quick convergence to the best solution, simple concepts, and ease of implementation [14]. These results inspired this study to further apply TLBO to the feature selection problem. However, TLBO similar to other approaches, there is also a chance of being caught in the local optimum under certain situations. Consequently, the original TLBO algorithm included two key improvements. The First improvement includes the use of binary modified teaching-learning-based optimization (BMTLBO) for feature selection. The transforming function converts the basic algorithm to the binary version, then for strengthening exploration capability a pooling concept is applied. This pool is formed by combining the worst solution with a promising solution, resulting in increased diversity. Along with strengthening exploitation in promising and nearly optimum areas, this concept is used to explore areas that have not yet been assessed. Using elitism helped the algorithm's iterations avoid losing potential points.

The Second improvement is, the using of a improved TLBO algorithm for calculating classification accuracy, the learner phase of the TLBO algorithm is improved with a self-learning concept (SLTLBO) to increase the exploitation capability, furthermore for increase exploration capability after every three sequential generations, if the learners do not improve, and they will start from a new random position. The SLTLBO is trained the neural network to improve the level of exploitation capability in the fundamental TLBO algorithm. Consequently, to fix the aforementioned problems with the fundamental TLBO algorithm, two improved TLBOs are proposed to enhance exploitation and exploration capability, BMTLBO is applied to the feature selection problem and SLTLBO for training artificial neural networks. In addition, the feature selection algorithms in this paper are an illustration of the wrapper-based category, which aims to improve classification accuracy.

The following contributions are recommended to fulfill this main goal: • BMTLBO: A more useable version of the basic algorithm that is binary in nature, with the basic algorithm's defects fixed, and adds a pooling mechanism to increase population diversity. In addition, the elitist method is also applied to prevent the elimination of potential points in the iterations of the algorithm. • SLTLBO: In this paper, a novel SLTLBO algorithm is presented and created. SLTLBO in the self-learning phase improves exploitation and also population diversity in the search process, to enhance the current best solution and overcome the local optima problem. • Utilizing 15 benchmark datasets from the UCI repository, this presented approach is contrasted with several other optimization algorithms, including (SSA, GA, PSO, GOA, and ALO). The comparison results show that the proposed method outperforms these optimization algorithms with the metric of classification accuracy, convergence speed, and feature reduction.

The rest of the essay is segregated into the following sections. In Section. 2, related work is reviewed. In Section. 3, the proposed method, BMTLBO and SLTLBO algorithm are described in detail. In Section. 4, the suggested method is evaluated with an illustration of the detailed result. Finally, in Sect. 5, the paper is concluded.

2. Related work

Big data has rapidly impacted all facets of human life over the past few years, big data has received a special popularity and the need for data mining is felt more than ever. One of the important tools in dealing with big data is feature selection, which along with reducing the dimension and removing extra and irrelevant features, increases the algorithm's efficiency and decreases the time complexity. This paper suggested evolutionary algorithms for feature selection. In the following, we will discuss the previous research in this field. Ramasamy et al., suggested a binary version of the modified Grey Wolf Optimizer (GWO) for feature selection to classify Parkinson's disease [15], additionally, a novel strategy is used to choose the wolf at random. On the other side to achieve the ideal subset of features for huge dimension data classification, the algorithm known as Trainable Stable GWO is recommended [16]. Mirjalili et al., suggested a novel Binary Dragonfly Algorithm (DA) that is categorized into wrapper-based feature selection methods, which was proposed for the classification of the Covid-19 datasets [17]. In another research the application of the Gravitational search algorithm (GSA) is clarified for feature selection [18], which has been enhanced by the genetic operators, including crossover and mutation. There is a hybrid technique for feature selection and clustering by Binary GSA [19]. Researchers have widely used meta-heuristic algorithms for feature selection, another algorithm in this field is Salp Swarm Algorithm. The Dynamic Salp Swarm Algorithm, an improved version of SSA, is presented by the authors to prevent stacking into local optima. This algorithm is combined with the K-nearest neighbor (KNN) classifier in a wrapper mode [20]. It demonstrates a salp swarm algorithm with two distinct methods, comparing the effectiveness of eight various transfer functions along with a crossover operator [21]. For the purpose of feature selection, an enhanced multi-objective Salp swarm algorithm is also developed [22]. Another researcher used this algorithm with an opposition-based learning population hybrid by local search algorithm to increase exploitation capability [23]. The best feature subset for classification purposes was determined by comparing various binary implementations of the Grasshopper Optimization Algorithm (GOA) [24]. The convergence of the fundamental algorithm has been improved through the use of a new combination method that uses the Evolutionary Population Dynamics, GOA and selection operators [25]. In another research Binary Dragonfly Algorithm (BDA) is applied for wrapper-based feature selection that focus on how to choose transfer functions [26]. WOA is a recently proposed algorithm that imitates the natural hunting strategy of humpback whales. To select the best feature subsets for classification, two binary variants of the WOA algorithm are suggested [27,28]. For detection and classification, an automatic feature

Table 1
Brief details about related works.

Reference	Year	Description Method	Type of Optimization	Classifier	Measures	Type of method
R. Ramasamy Rajammal, S. Mirjalili, G.Ekambaram, N.Palanisamy [15]	2022	Binary Grey Wolf Optimizer	Single objective	KNN	*Classification Accuracy	Wrapper
J. Too and S. Mirjalili [17]	2020	Hyper Learning Binary Dragonfly Algorithm	Single objective	KNN	*Classification Accuracy	Wrapper
M. Taradeh [18]	2019	Gravitational Search	Single objective	KNN	*Classification Accuracy *Number of Features	Wrapper
R. Guha, M. Ghosh, A. Chakrabarti, R. Sarkar, and S. Mirjalili [19]	2020	Clustering Based Binary Gravitational Search Algorithm	Single objective	KNN	*Classification Accuracy *Number of Features *K_Value	Wrapper
M. Tubishat et al. [20],	2021	Dynamic Salp Swarm Algorithm (DSSA)	Single objective	KNN	*Classification Accuracy *Number of Features	Wrapper
I. Aljarah et al. [22],	2020	Multi-Objective Salp Swarm Algorithm (MODSSA-Lbest)	Multi-objective	KNN	*Error rate *G-mean (P-values) *Recall *Specificity *Number of Features	Wrapper
M. Tubishat, N. Idris, L. Shuib, M. A. M. Abushariah, and S. Mirjalili [23]	2020	Improved Salp Swarm Algorithm (ISSA)	Single objective	KNN	*Classification Accuracy *Number of Features *fitness (MSE)	Wrapper
M. Mafarja and S. Mirjalili [27],	2018	Whale Optimization Algorithm	Single objective	KNN	*Average Selection Ratio *Classification Accuracy *Time complexity	Wrapper
M. Mafarja and S. Mirjalili [28],	2017	Hybrid Whale Optimization With Simulated Annealing	Single objective	KNN	*Classification Accuracy *Number of Features	Wrapper
M. S. Abbasi, H. Al-Sahaf, M. Mansoori, and I. Welch [29]	2020	Particle Swarm Optimization	Single objective	RLR, RF, DT, KNN, SVM	*Classification Accuracy *Number of Features	Wrapper
W. Xie, L. Wang, K. Yu, T. Shi, and W. Li [31]	2023	Binary Firefly Algorithm (Mrmr-MBFA)	Single objective	KNN	*Classification Accuracy *Number of Features	Wrapper
M. Allam and M. Nandhini [33]	2022	Binary Teaching Learning Based Optimization (Fs-Btlbo) Algorithm	Single objective	SVM	*Classification Accuracy *Number of Features	Wrapper
S. Thawkar [34]	2021	Hybrid Teaching-Learning-Based Single Objective Optimization And Salp Swarm Algorithm (TLBO_SSA)	Single objective	ANN	*Accuracy *Recall *Sensitivity *Specificity *False Positive Rate *False Negative Rate *F_Measure *Cohn's Kappa Coefficient	Wrapper

selection technique based on particle swarm optimization is used [29]. Optimal feature evaluation and selection (OFES), a method used to evaluate and select high-quality features for multi-class classification, is a fascinating study on multiclass problems [30]. Two phases of an enhanced multilayer binary firefly-based method are proposed, Enhanced Min-Redundancy and Max-Relevance in the first stage reduce the dimensionality of the space, at each iterative recursive step, the improved binary firefly algorithm emend the feature space from coarse to fine-grained level [31]. TLBO algorithm is a relatively new, very widely used, low-parameter and powerful in solving complex problems that possess a quick convergence rate.

In the following, we mention some applications of this algorithm in feature selection, The Chebyshev distance formula is used in an improved algorithm based on training and learning that looks to identify the ideal feature subset [32]. In the development of the binary TLBO algorithm, only a few widely used controlling variables, like population size and the number of generations, can be used to extract the best features subset from a dataset [33]. This algorithm performs remarkably well when combined with other algorithms, such as when SSA is used for feature selection and an artificial neural network is used as the classifier. An adaptive neuro-fuzzy inference system assesses the obtained model [34]. A brief summary of related works is provided in Table 1.

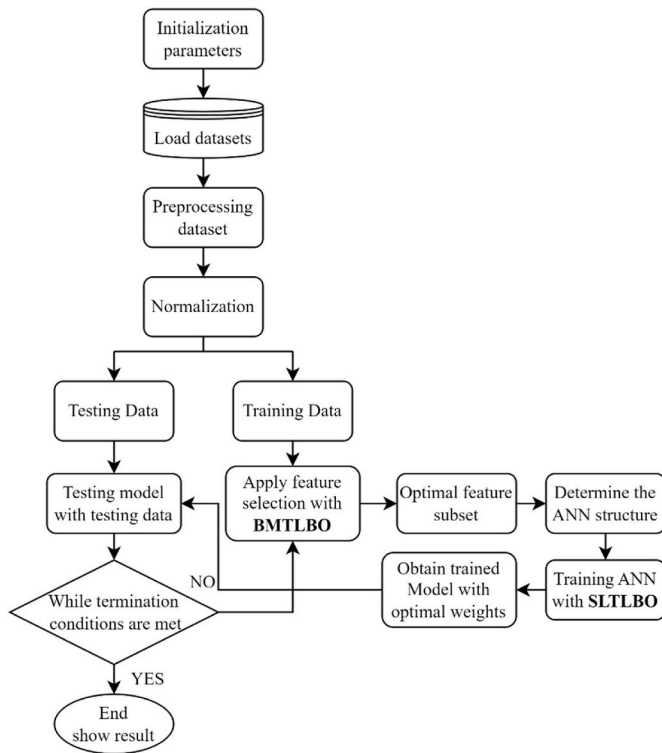


Fig. 1. The flowchart of our proposed model.

3. Proposed model

A population-based heuristic algorithm with excellent capabilities for resolving complex issues is the TLBO algorithm. This algorithm has straightforward implementation and execution, straightforward concepts, few parameters (requires determining the population size and the number of iterations there will be), strong power of exploration and flexibility. This algorithm can indeed generate optimal solutions in a reasonable time, but besides all these benefits in handling some complex problems, there is a possibility of convergence to the local optimum. This section presents the two modified TLBO algorithms. First, the BMTLBO algorithm is applied for feature selection to decrease the computational cost and improve the accuracy and then SLTLBO is used for calculating classification accuracy. Classification accuracy means the prediction accuracy of the model.

Given the benefits of the TLBO algorithm that were previously mentioned and its superiority over other optimization techniques, BMTLBO algorithm with strengthen exploration capability using pooling mechanism is used for feature selection problem and another improved version of it called SLTLBO has been used to train the neural network and determine the classification accuracy. By incorporating the ideas of neighborhood and future self-learning into the fundamental steps of the basic algorithm, this algorithm strengthens its exploitation power while maintaining a balance between exploration and exploitation. Because models trained using this method are capable of accurately classifying high-dimensional data, the improved version of SLTLBO will be a useful technique for neural network training. The general configuration of the method is shown in Fig. 1.

As shown in Fig. 1 first we initialize parameters and then after applying a preprocessing phase (for example converting the nominal value to the numerical) on datasets, data are normalized, then the BMTLBO algorithm starts the feature selection process. In BMTLBO by using a pooling operator, its population diversity is improved and avoids it from stuck into local optima. This proposed BMTLBO algorithm works with a neural network classifier using wrapper mode for feature selection problems. SLTLBO is developed for training neural networks as a

classifier to calculate algorithms performance. Modifications are applied to this algorithm to avoid trapping stuck in local optima. In addition, by adding the self-learning phase and an operation that preserves the diversity in the population, the exploration and exploitation capability of the algorithm is strengthened, so a balance between them is established. Fig. 1 shows the general framework of the method. For each of these algorithms, a detailed explanation will be presented in the next subsections.

3.1. Teaching learning-based optimization (TLBO) algorithm

A population-based meta-heuristic algorithm that inspired the teaching and learning processes in a classroom [13]. One of the most effective and appealing optimization strategies that have emerged is the TLBO algorithm. It has been proved that the TLBO algorithm is a powerful optimization technic and has a quick convergence rate. However, it is possible to become trapped in a local optimum. To enhance the basic TLBO algorithm's capacity for exploitation and exploration, numerous modifications and extensions have been implemented [35]. Since there are fewer tuning parameters for the TLBO algorithm, it is simpler to use. The Teacher Phase and Learner Phase are the two phases of this algorithm. Teachers make an effort to raise their students' knowledge levels, and students also try to learn from each other lowing provides a brief explanation of two main steps of the algorithm.

3.1.1. Teacher phase

In this stage, the classroom teaching procedure is demonstrated. According to each student's capacity, the teacher makes an effort to raise the students' knowledge level. The positions of the learners are updated during this phase using Equation. 1.

$$X_{i,new} = X_{i,old} + r \cdot (X_{teacher} - T_F \cdot X_{mean}) \quad (1)$$

Where $X_{teacher}$ is the teacher with the best value of fitness function, X_{mean} is the average value of the whole population and r is a number that is chosen at random range of (0, 1), as evidenced by the literature, T_F is a teaching factor. TLBO performs better when T_F is set to a value between 1 and 2. If T_F is set to 1, the teacher has the biggest impact on the learner's new position. The new position of learners will have the highest level of diversity when T_F is increased by 2. Following is the calculation for the random selection of T_F .

$$T_F = \text{round} [1 + \text{rand}(0, 1)] \quad (2)$$

Moreover, X_{new} is the updated value of X_i in each iteration of the algorithm [36]. If the updated position's function value is superior to the old positions, it will be accepted.

3.1.2. Learner phase

Interaction between students helps them learn more during this step. The mutual interaction process generally raises the learner's grade. The random interaction among learners improves their knowledge. If the randomly chosen learner X_j has more knowledge than the selected learner X_i , the X_i will receive new information. To further his or her understanding of Equations. 3 and 4, a learner randomly interacts with other learners

$$X_{i,new} = X_{i,old} + r \cdot (X_j - X_{i,old}) \quad \text{if } f(X_i) > f(X_j) \quad (3)$$

$$X_{i,new} = X_{i,old} + r \cdot (X_{i,old} - X_j) \quad \text{if } f(X_i) < f(X_j) \quad (4)$$

3.2. Binary modified teaching-learning based optimization (BMTLBO) algorithm

In this section, with the aim of strengthening the local and global search of the basic algorithm and improving the diversity of the population, we have developed the basic TLBO algorithm for feature

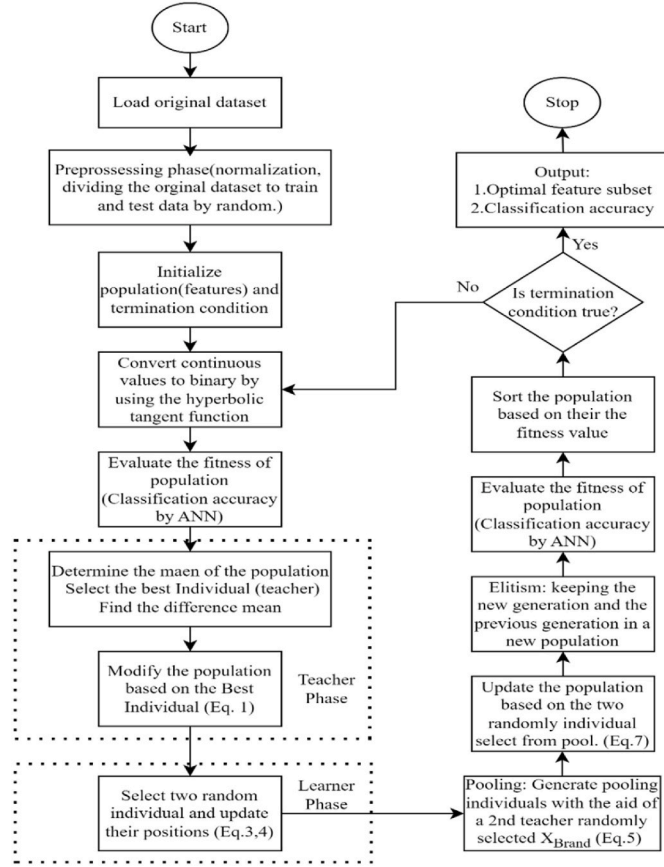


Fig. 2. The BMTLBO algorithm for feature selection.

The pseudo code for BMTLBO	
1.	Initialization parameters (Objective function $f(X)$ $X = (x_1, x_2, \dots, x_d)^T$)
2.	Generate initial learner position and calculate objective function $f(X)$ for them
3.	While (the termination conditions are not met)
4.	%%Teacher Phase
5.	Calculate the mean of each design variable X_{mean}
6.	Identify the best solution ($X_{teacher}$)
7.	For $i = 1 \rightarrow \text{pop size}$
8.	Calculate teaching factor T_f using Equation. (2)
9.	Modify solution based on best solution (teacher) using Equation. (1)
10.	%%Learner Phase
11.	Randomly select two learners, and update according to Equation. (3,4).
12.	End for
13.	Calculate the second best position randomly selected from around the teacher (Equation.6)
14.	Pooling : Filling the pool with Equation. (5)
15.	For $i = 1 \rightarrow \text{pop size}$
16.	Modify the solution based on two positions randomly selected from the pool (Equation.7)
17.	End for
18.	%%Elitism Phase
19.	New Population = [previous generation, current generation];
20.	Covert continues values to the binary values by Equation. (8)
21.	Calculate objective function $f(X_{new})$
22.	Sort the New Population based on objective functions value.
23.	Keeping the top population as the number of pop size
24.	Set iteration = iteration + 1
25.	End while
26.	Post-process results and visualization

Fig. 3. The BMTLBO algorithm pseudo code.

selection. In this section, a modified version of the TLBO method is proposed. This algorithm is developed to enhance the accuracy and convergence rate of the standard TLBO. The standard algorithm is modified in two general steps. The main parts of BTLBO are as follows.

Step 1 (Elitism): The basic algorithm has two selection operators during the training and learning phase that accept the new individual if they improve and immediately remove the previous individual from the population. This unequivocal removal of population individuals may trap the algorithm in the local optimum in subsequent iterations because

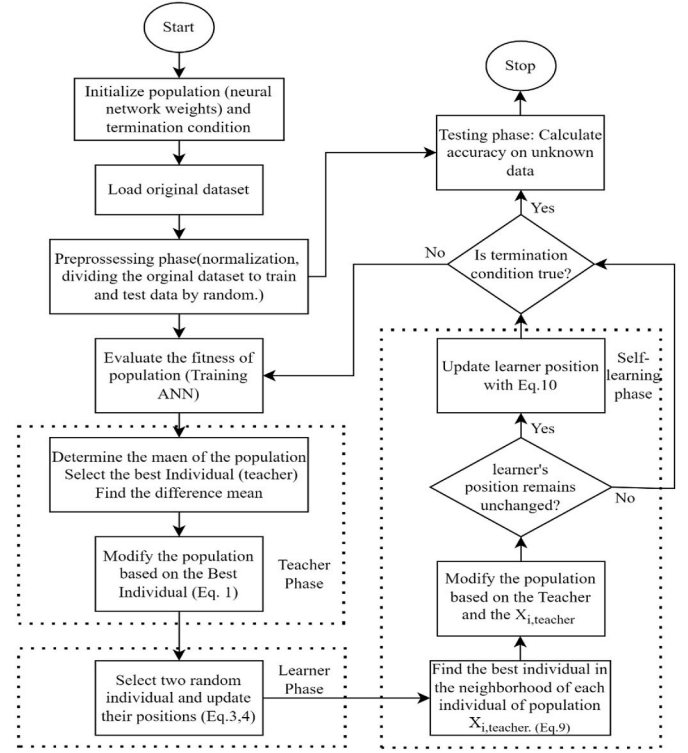


Fig. 4. The SLTLBO algorithm for training a neural network.

The pseudo-code for SLTLBO	
1.	Initialization parameters (Objective function $f(X)$ $X = (x_1, x_2, \dots, x_d)^T$)
2.	Generate initial learner position and calculate objective function $f(X)$ for them
3.	While (the termination conditions are not met)
4.	Calculate the mean of each design variable X_{mean} %%Teacher Phase
5.	Identify the best solution ($X_{teacher}$)
6.	For $i = 1 \rightarrow \text{pop size}$
7.	Calculate teaching factor T_f using Equation. (2);
8.	Modify solution based on best solution (teacher) using Equation. (1);
9.	Calculate objective function for new mapped student $f(X_{new})$
10.	IF X_{new} is better than X_i
11.	$X_i = X_{new}$
12.	End if
13.	IF $\text{rand} < \text{Pc}$ %%Learner Phase
14.	Find the best learner around her/his, and update according to Equation. (9)
15.	Else
16.	Randomly select another learner X_j , and update according to Equation. (3), (4).
17.	End if
18.	IF individual's position didn't change after some iteration
19.	The position is updated according to Equation. (10).
20.	End if
21.	IF new positions are better than the Previous position replace them
22.	End if %%End of Learner Phase
23.	IF $\text{mod}(\text{iteration}, m) = 0$
24.	randomly rearrange all individuals
25.	End if
26.	End for
27.	Set iteration = iteration + 1
28.	End while
29.	Post-process results and visualization

Fig. 5. The SLTLBO algorithm pseudo code.

the removed individual may perform better than other population individuals in other iterations of the algorithm in combination with the best individual or individuals close to the global optimum. Therefore, it is suggested to keep the current and next population individuals for the entire steps of the algorithm. Then the individuals are sorted based on their evaluation function value and the first Pop Size individuals which contain the lowest cost function value (Pop Size is the number of population individuals) are used for the next iteration of the algorithm and the optimization process continues with the new population obtained. This operation increases the accuracy and convergence rate of the original TLBO because valuable individuals of the population are not

Table 2
Datasets description.

	Dataset		Number of features	Number of instances
1	Nsl-kdd	Train data	42	125974
		Test data	42	22545
2	phishing		30	2457
3	ionosphere		34	351
4	credit		20	1000
5	spambase		57	4601
6	heart		13	270
7	lymphography		18	148
8	spect		22	267
9	vote		16	300
10	Australian		14	690
11	dermatology		34	366
12	Satellite		36	6435
13	waveform		21	5000
14	sonar		60	208
15	Biodegradable		41	1055

Table 3
Comparison between BMTLBO and TLBO based on average accuracy and average-number-of-selected features.

Dataset	Average classification accuracy		Average Number of selected feature	
	BMTLBO	TLBO	BMTLBO	TLBO
Nsl-kdd	97.4	95.9	27.5	29.4
phishing	97.3	97.4	18.3	20.7
ionosphere	97.5	96.3	10.9	14.4
credit	80.9	78.3	7.9	9.5
spambase	97.6	93.5	25.3	27.4
heart	91.3	85.8	5.9	7.3
lymphography	85.4	71.3	8.2	10.5
spect	85.9	77.6	8.9	11.2
vote	98.7	98.4	5.7	7.9
Australian	89.9	86.5	5.1	8.3
dermatology	97.6	92.4	12.6	16.7
Satellite	94.6	89.1	17.5	19.5
waveform	82.1	80.3	11.5	13.8
sonar	98.9	96.4	15.7	24.8
Biodegradable	87.7	84.2	10.5	18.6

necessarily removed in the initial generations.

Step 2 (Pooling): This improvement to the standard TLBO algorithm is motivated by the idea that, in the classroom, when there is no teacher present, students are not in the class, and the school is closed, they approach the best student and attempt to solve their problems. In other words, it can be noted that people who are not very close to the global optimum and do not have a promising position may have a better result if they interact with an individual whose position is close to the optimum. A pool, whose individuals are valued as follows, is considered to scour unvisited areas, increase population diversity, and boost exploitation in promising areas. This pool consists of K number individuals with the dimensions of different problems.

$$P_j = B_j \times X_{Brnd} + \bar{B}_j \times X_{wrost} \quad j = 1, \dots, K \quad (5)$$

Where B_j is a binary random vector and $\bar{B}_j$ has the reverse value of B_j such that the zero values in B_j are equivalent to 1 in $\bar{B}_j$, and the values of non-zero in B_j are equivalent to the corresponding zero in $\bar{B}_j$, X_{wrost} is equal to the position of the worst individual of the population and X_{Brnd} is calculated by Equation (6).

$$X_{Brnd} = rand \times (up_{teacher} - lb_{teacher}) + lb_{teacher} \quad (6)$$

Where X_{Brnd} is a promising individual around the Teacher (best individual) from the population that is selected randomly, $rand$ is a random number from (0, 1) interval and $up_{teacher}$, $lb_{teacher}$ are the upper bound and lower bound of Teacher's position, respectively. Since the good

student in different courses may not always be the second-best individual in the population, we randomly select an individual from the population who is close to the teacher's position. Finally the population will be updated by Equation. 7.

$$X_{i,new} = X_{i,old} + rand \times (P_{1rnd} - P_{2rnd}) \quad (7)$$

Where $X_{i,new}$, $X_{i,old}$ are the position of X_i in the current and previous generation, respectively and P_{1rnd} , P_{2rnd} are two individuals that randomly selected from the pool.

One of the most practical big data problems is feature selection. The feature selection algorithm is proposed as a binary modified teaching learning algorithm. The learner's positions, which are produced from the TLBO algorithm, are continuous values. Continuous values must first be converted to binary using a function because the feature selection problem is binary in nature and cannot be solved using continuous values. A transformation function performs the mapping from the continuous search space. The function of the hyperbolic tangent is used in the mapping.

$$V = |\tanh(x_i)|, x_{binary} = \begin{cases} 0 & \text{if } rand < V \\ 1 & \text{if } rand \geq V \end{cases} \quad (8)$$

Where x_i is a continuous value and x_{binary} value can be 0 or 1 depending on the value of a random number, the method can transform the continuous search space into a binary one.

BMTLBO a novel wrapper method: For the feature selection problem, the proposed BMTLBO algorithm is applied to a neural network classifier in wrapper mode. In each iteration, the training dataset will be run through BMTLBO to determine the subset of features that will be used to train the ANN classifier. Selected and unselected features in a feature selection problem are represented by binary values, with the "1" value in the solution specifying a selected feature and the "0" value specifying an unselected one. The steps of the proposed BMTLBO algorithm are as follows:

- **BMTLBO initial population:** The population is randomly initialized. According to the value of the population size parameter, a subset of features is chosen for each individual of the population.
- **Produce new generation:** BMTLBO updates the positions of each learner using teacher phase and learner phase Equations (1)–(4) and finally, we apply the hyperbolic tangent function to transform continuous values into binary values.
- **Fitness evaluation:** for calculating fitness, the SLTLBO algorithm will determine the fitness values of all learners by training a neural network. The fitness value is evaluated by the neural network that has been trained by the SLTLBO algorithm. Then BMTLBO will update the value of learners if there is a better solution.
- **Pooling:** The pooling mechanism is used to enhance algorithm exploration while preserving population diversity. The value of pooling individual positions is updated by the result of the worst individual in the population and the better person who is randomly selected from around the best individual in the population. With this improvement Accuracy will be increased, the convergence process will be improved, and the local optimum will not become stuck in the local optimum. This method's diversity greatly reduces the chance of getting stuck in the local optimum.
- **Elitism:** One of the weaknesses of the fundamental TLBO algorithm is that, after updating the positions of each population individual, if the current position is better than the previous position of the same individual, the previous position is simply deleted. However, it's possible that in the same algorithm iteration, the deleted position is superior to other ones. So, all of the individuals of the current generation and the previous generation are kept in this section, and they are sorted according to the fitness function value at the end of the algorithm's iterations. In the end, we will retain better individuals based on population size.

Table 4

Comparison between BMTLBO and other evolutionary algorithms based on average accuracy and average-number-of-selected features.

Dataset	Metrics	BMTLBO	SSA	GOA	PSO	ALO	GA
Nsl-kdd	Accuracy	97.4	93.6	92.1	95.7	96.4	94.9
	# Features	27.5	25.9	28.5	29.3	30.4	27.4
phishing	Accuracy	97.3	96.4	94.8	87.3	89.8	88.9
	# Features	18.3	19.6	21.8	19.2	25.4	20.7
ionosphere	Accuracy	97.5	97.1	92.8	97.4	93.7	96.5
	# Features	10.9	13.9	15.7	15.1	16.4	11.7
credit	Accuracy	80.9	74.6	71.5	75.4	69.1	76.4
	# Features	7.9	9.4	10.0	9.7	10.7	9.3
spambase	Accuracy	97.6	93.4	90.6	93.8	90.4	93.4
	# Features	25.3	26.3	28.7	30.1	29.5	22.2
heart	Accuracy	91.3	85.9	78.8	86.2	75.5	84.4
	# Features	5.9	5.6	6.4	6.2	5.9	5.4
lymphography	Accuracy	85.4	68.1	54.8	72.2	54.5	68.5
	# Features	8.2	7.9	8.2	8.5	8.7	8.2
spect	Accuracy	85.9	81.6	70.4	81.6	72.6	81.3
	# Features	8.9	10.5	10.8	10.0	11.2	10.5
vote	Accuracy	98.7	98.3	96.2	97.3	96.3	97.6
	# Features	5.7	6.2	7.7	7.4	7.6	3.5
Australian	Accuracy	89.9	84.0	78.8	83.7	76.0	84.3
	# Features	5.1	5.9	5.9	6.9	6.7	5.2
dermatology	Accuracy	97.6	99.1	98.7	98.8	95.1	96.7
	# Features	12.6	15	16.8	15.9	16.5	14.4
Satellite	Accuracy	94.6	91.8	90.5	92.5	90.5	92.1
	# Features	17.5	19.7	18.7	19.3	18.3	22.6
waveform	Accuracy	82.1	80.2	76.5	80.0	77.4	79.8
	# Features	11.5	12.6	11.3	11.5	11.6	13.0
sonar	Accuracy	98.9	96.1	90.3	96.6	90.3	97.5
	# Features	15.7	26.4	29.6	26.9	29.3	22.6
Biodegradable	Accuracy	87.7	85.1	87.2	84.3	85.4	86.9
	# Features	10.5	23.8	24.7	23.1	16.3	17.2

- Optimal feature subset: The best subset of features that BMTLBO has selected will be represented by the best solution that is returned by BMTLBO.
- Testing section: The features that are selected as the best solution will be used in this phase to assess how well SLTLBO performs on the testing dataset. Figs. 2 and 3 describe in each step how BMTLBO functions

3.3. Self-learning teaching-learning based optimization (SLTLBO)

For neural network training, an improved teaching-learning algorithm is proposed (Figs. 4 and 5). TLBO is a strong algorithm that can achieve high-quality results and has excellent convergence stability [5], but there is a chance of becoming trapped in the local optimum. The learning phase with the concept of neighborhood is proposed in this improved algorithm to prevent the algorithm from incoming the local optimum. In this manner, the algorithm tries to decrease random choices and use the best neighbors of each person for learning to increase the local and global search. The main steps of SLTLBO are as follows:

3.3.1. Self-learning phase

Along with the teacher's expertise, each student in the class can learn from their classmates. So learners may receive tutoring from their peers or top learners in the class, so in each classroom, they can learn from the best learner around them. To strike a balance between exploration and exploitation, the neighborhood concept is used to enhance the algorithm's convergence process. The order in which learners are seated in the classroom is messed up after several iterations of the algorithm to preserve diversity. Equation. 9 is used to update the learners' positions during the self-learning phase.

$$X_{i,new} = X_{i,old} + r_2 \cdot (X_{i,teacher} - X_{i,old}) + r_3 \cdot (X_{i,teacher} - X_{i,old}) \quad (9)$$

Where $X_{i,teacher}$ is the best learner in $X_{i,old}$ neighborhood, $X_{i,teacher}$ is the teacher, r_2 , r_3 are random numbers in (0, 1) interval. To improve a learner's ability to search, people update their positions by the current

teacher and the mean of the learners if a learner's position doesn't change over several iterative generations. The update method is put into practice as follows:

$$X_{i,new}(t) = \text{normrnd}\left(\left(\frac{X_{i,teacher}(t) + X_{i,mean}(t)}{2}\right), \text{abs}(X_{i,teacher}(t) - r_5 * X_{i,mean}(t))\right) \quad (10)$$

Where r_5 when the position of the teacher is equal to the mean position of the students, increases performance by the distribution of new individuals.

4. Experiment

4.1. Description of datasets

The proposed method's effectiveness was examined using 14 benchmark datasets from the UCI dataset repository to compare it to other methods, assess its merits, and demonstrate its superiority over existing approaches. Information about the used datasets is provided in Table 2. Additionally, The effectiveness of the suggested method on chemicals is tested using a set of chemical data (QSAR biodegradation Data Set) with the help of this dataset, researchers have been able to better understand the connections between molecular biodegradation and chemical structure. The data are divided into two categories by the target values in the final column: biodegradable (RB) materials and non-readily biodegradable (NRB) materials. Normalization data is consist of several step, • it is possible to be some features that resemble strings in the dataset (for example NSL-KDD dataset). They must be transformed into numbers because the suggested model relies on numbers to function. Additionally, it decreases the amount of data, which uses less memory.

- There are some data classes where there aren't enough training examples, and the model can't properly learn them. These data do not have to be kept in the data set.
- Dataset scaling is the next step. All the data values are converted to the range [- 1, 1], because deep learning

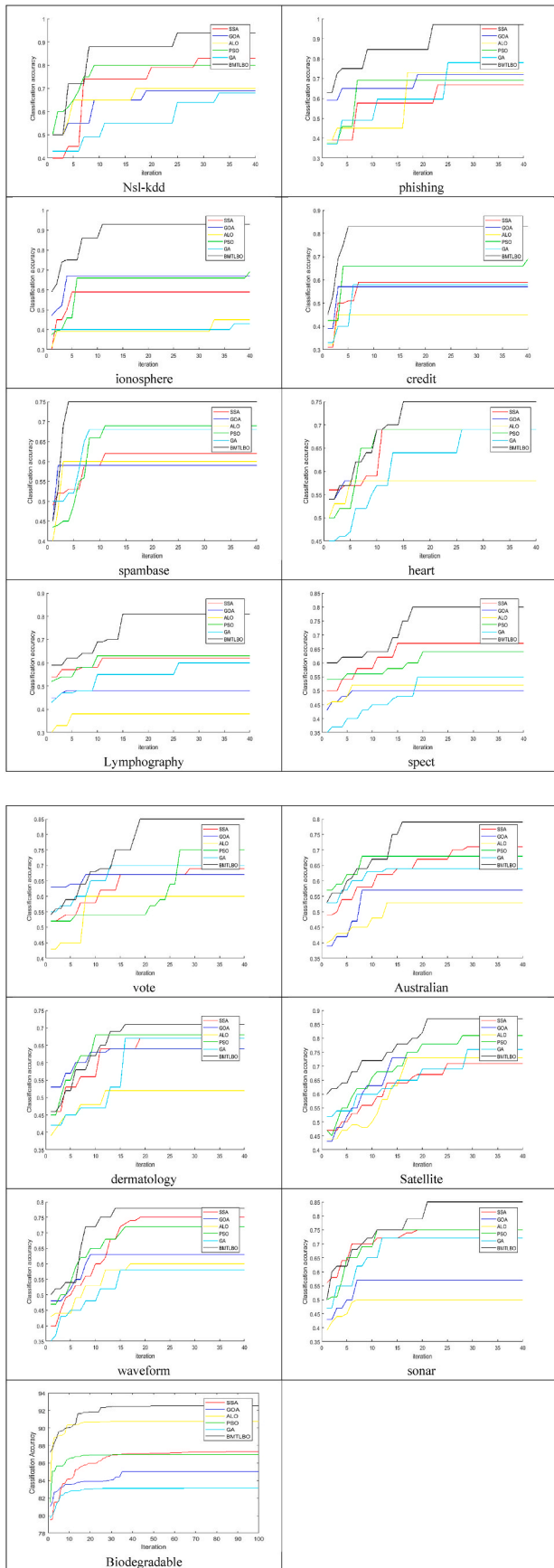


Fig. 6. The Convergence curve of our proposed model compared to other algorithm.

models perform better with smaller amounts of data. Data are computed after normalization making use of the MapMinMax algorithm.

4.2. Parameter settings

The overall average result has been reported for each experiment that has been conducted. The outcomes in Table 2 represent the average values over 20 runs in terms of classification accuracy and the number of selected features. The proposed algorithm is compared to other optimization algorithms like the GA, PSO, ALO and GOA, as well as the standard TLBO. In addition, a maximum of 100 iterations are allowed for each optimization algorithm, and a population size of 50 is used for all algorithms. The multi-layer perceptron (MLP) neural network is used as the classifier, with classification accuracy serving as the primary performance indicator for the optimization algorithms. The proposed method was implemented and evaluated in MATLAB R2017b on a Windows computer with an Intel® Core™ i5 CPU and a speed of 2.30 GHz and 8 GB of RAM.

4.3. Result and analysis

This section displays the particulars and results of each experiment. In the first experiment, the proposed BMTLBO is compared to the standard TLBO. The second experiment contrasts the performance of the BMTLBO algorithm with that of the GA, PSO, GOA, and ALO algorithms. Two metrics are used to compare BMTLBO and other algorithms: classification accuracy and the number of features that are selected. In this paper, the term “classification accuracy” refers to the model’s prediction accuracy, or the model’s precision when applied to unreliable data.

4.3.1. Comparison of BMTLBO and TLBO

The suggested model is contrasted with the fundamental TLBO algorithm in this section. It is unequivocally shown that BMTLBO significantly outperforms the original TLBO algorithm in terms of performance. The classification accuracy of both algorithms for all 15 datasets is presented in Table 3, which demonstrates that the BMTLBO algorithm performs significantly better than the standard TLBO algorithm. Additionally, because BMTLBO selects fewer features across all 15 datasets than TLBO, our method outperformed it in terms of the number of features, as shown in Table 3. By comparing the reported results to the original TLBO algorithm, it is evident that BMTLBO performs significantly better.

4.3.2. Comparison of BMTLBO with other optimization algorithms

The superiority of our proposed model is illustrated using a variety of optimization algorithms. BMTLBO and TLBO are compared in the previous section, the results showed that BMTLBO performed better than the TLBO algorithm’s standard version. Our proposed algorithm is compared to other optimization algorithms to demonstrate its superiority even more. In terms of average classification accuracy and average number of features chosen, our suggested model is compared. BMTLBO has been compared with Salp Swarm Algorithm (SSA) [37], Grasshopper Optimization Algorithm (GOA) [38], Particle Swarm Optimization (PSO), Ant-Lion Optimization (ALO) [39] and Genetic Algorithm. Comparative results between BMTLBO and other algorithms are shown in Table 4. In terms of classification accuracy, Table 4 clearly shows that BMTLBO performed better than all other algorithms, but the SSA algorithm performed better in the dermatology dataset. In 9 data sets, the proposed model attempts to select the fewest number of features possible, which lowers the method’s time complexity. As a result, the average number of features selected is lower. Table 4 shows that the GA algorithm can handle fewer features than other techniques in 4 data sets and also SSA was able to choose fewer features. In the case of the NSL-KDD dataset. To demonstrate our method’s superiority over other algorithms, its convergence curve is compared to those of other algorithms in the following (see Fig. 6).

According to Fig. 6, our approach has a higher capacity for exploration and exploitation, which, in addition to preserving diversity, leads to a more optimal search of the feature space. The results of our method typically outperform those of other algorithms, as shown by the convergence curves. This is evidence that it is possible to avoid becoming trapped in the local optimum. Other algorithms, however, are easily snared by the local optimal trap, as shown by the Figure and the outcomes.

5. Conclusion

This paper suggests the simultaneous use of two novel TLBO algorithms, BMTLBO and SLTLBO, for feature selection and neural network training, respectively. The two improvements are used to balance the capacity for exploration and exploitation and prevent trapping in local optima. For strengthen the Teacher phase of the standard TLBO in the case of exploration capability, a pooling concept is added to the BMTLBO algorithm to improve the level of population diversity. Additionally, a self-learning component is applied to the SLTLBO algorithm for a more effective search of the search space. Over 14 datasets from the UCI repository, the suggested method is tested and assessed. A chemical data set called Biodegradable is also used to test and evaluate the proposed method in order to demonstrate its use in the field of chemistry. Based on classification accuracy and the number of selected features, our proposed model is compared to several well-known optimization algorithms (SSA, GA, PSO, GOA, and ALO). Through several experiments, the superiority of the proposed approach to other algorithms is shown. Future studies could examine the application of the TLBO algorithm to problems like sentiment analysis for disease diagnosis, cancer detection, and intrusion detection systems.

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Declaration of competing interest

Conflict of Interest: The authors declare that they have no conflict of interest.

Data availability

No data was used for the research described in the article.

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